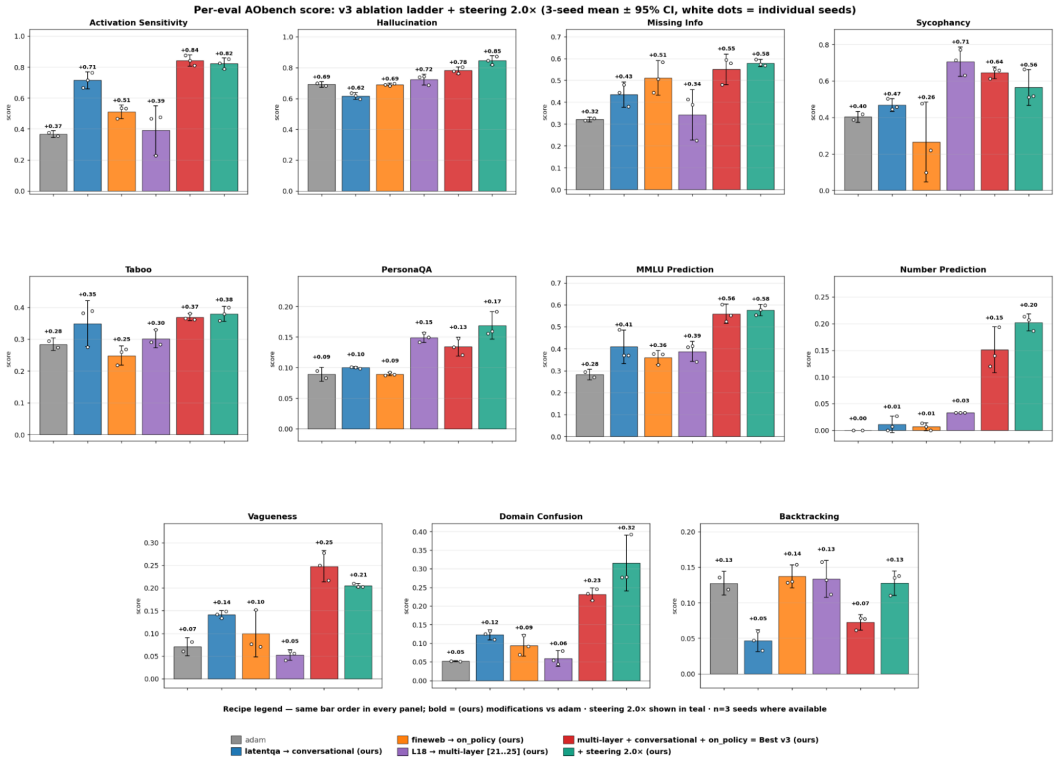


Figure 12: **Per-eval AObench scores across the ablation ladder.** Bars show chance-adjusted scores per task for each recipe in our ablation; white dots are individual seeds. Higher is better for every eval. The broad uplift visible in Fig. 7 is reflected across the majority of individual tasks, with no single task driving the result.

A.6 AObench details

AObench comprises the following individual evaluations, each chosen to target a capability an ideal AO should possess while attempting to remain robust to text-inversion confounds:

- **Activation Sensitivity** ("Not Just Reading Tokens"). Feeds the exact same set of tokens but sourced from different upstream contexts, and checks whether the AO produces meaningfully different answers. This provides a relatively clean test of whether the AO relies on activation-specific information rather than surface token patterns.
- **Hallucination** ("Not Obviously Wrong"). Checks whether the AO is obviously confabulating given the context the activation comes from, i.e. whether it invents specific but unsupported details about the model's reasoning.
- **Missing Information** ("Detect Missing Info"). Adapted from Jakkli et al. [2026], using SimpleBench-style questions where the model lacks a key piece of information but does not recognise this gap. Tests whether the oracle can surface knowledge that is not explicitly present in the reasoning.
- **Sycophancy**. Based on the Scruples dataset, this task assesses whether the oracle can detect sycophantic responses from the underlying activations produced during generation.
- **Taboo** ("Detect Taboo"). A model-organism task derived from Cywiński et al. [2025] and reused from Karvonen et al. [2025]: the oracle must identify a secret word that the model deliberately avoids mentioning.

[7] A thing we experimented with was trying to do 2 epochs: 1 where the data is 90% future/past-lens and 10% conversational, to teach capability, and then a second epoch which was 90% conversational, and 10% future/past-lens, to hopefully make it answer more conversational, thus increasing specificity/reducing hallucinations. This did not lead to meaningful uplift.

[8] Our chain-of-thought corpus consisted disproportionately of math, which is probably not optimal.

- **PersonaQA** ("Identify Persona"). Questions inquiring about the system prompt in use, approximating a steering vector; reused from Karvonen et al. [2025]. Tests the oracle's knowledge of the instructions shaping the model's behaviour.
- **MMLU Prediction**. Requires predicting the answer the model would give to an MMLU question if it were interrupted mid-reasoning and forced to respond immediately.
- **Number Prediction** ("Predict Hidden Number"). Predicts the numbers involved in an arithmetic equation from the activations of that equation, testing numerical extraction.
- **Vagueness** (inverse of "Response Specificity"). Measures whether the oracle's description of the model's reasoning is concrete, problem-specific and falsifiable, as opposed to a generic, unfalsifiable statement.
- **Domain Confusion** ("Identify Problem Domain"). Measures how often the model is completely wrong about the domain the activation is from (e.g. confusing ice cubes with hen houses), testing basic contextual awareness.
- **Backtracking**. Predicts whether the model is about to revise its current line of reasoning, which requires reading internal uncertainty signals.

The hallucination score reported in §4 is a normalised average of: "Not Obviously Wrong", "Identify Problem Domain", "Detect Missing Info", and "Predict Hidden Number". Vagueness in §4 is the inverse of "Response Specificity".

"Not Just Reading Tokens" is a particularly promising eval, where we observe significant uplift: it feeds the exact same set of tokens but in different upstream contexts and checks whether the AO produces meaningfully different answers, providing a relatively clean test of whether the AO uses activation-specific information. "Identify Persona" and "Detect Taboo" are taken from Karvonen et al. [2025].

Our evaluation tasks are open-sourced at cot-oracle. We reiterate that evaluating AOs is hard, chiefly due to the need to control for text inversion, and that judging vagueness reliably requires careful prompt design and manual spot-checking. We recommend qualitative analysis of every new AO checkpoint in addition to AObench.